



《编译原理》

第2讲: OCaml简介

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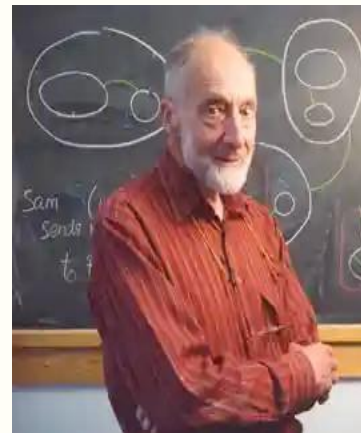
OCaml

```
# brew install opam  
# opam init  
# opam switch create ocaml-latest
```

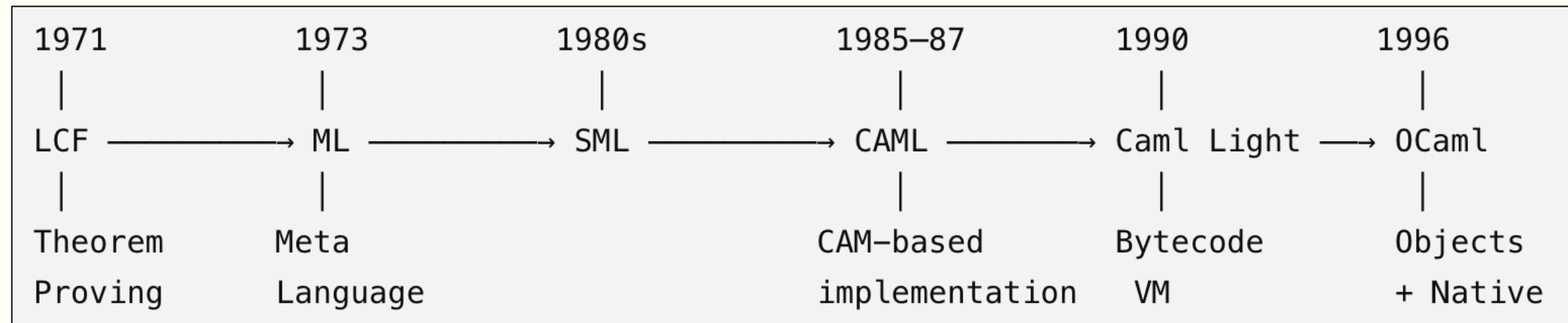
<https://ocaml.org/docs/installing-ocaml>

❑ OCaml: Objective Categorical Abstract Machine Language

❑ A brief history



Robin Milner



Xavier Leroy

❑ OCaml is a general-purpose, strongly typed programming language

- successor of Caml Light (itself successor of Caml)
- part of the ML family (SML, F#, etc.)

❑ Designed and implemented by **Xavier Leroy** and others at Inria Rocquencourt, France

❑ Some applications:

- symbolic computation and languages (IBM, Intel, Dassault Syst`emes),
- static analysis (Microsoft, ENS), file synchronization (Unison), peer-to-peer (MLDonkey), finance (LexiFi, Jane Street Capital), teaching

First steps with OCaml

□ The first program

hello.ml

```
print_string "hello world!\n"
```

compiling

```
% ocamlpt -o hello hello.ml
```

executing

```
% ./hello  
hello world!
```

Variables and References

❑ `let x = e` introduces a global **variable**

❑ Differences with respect to usual notion of variable:

- necessarily **initialized**
- type not declared but **inferred**
- **cannot be assigned**

Java

`final int x = 42;`

OCaml

`let x = 42`

❑ A variable to be assigned is called a **reference**

- it is introduced with **ref**

```
let x = ref 1;;  
print_int !x;;  
x := !x + 1;;  
print_int !x;;
```

Expressions and Statements

❑ no distinction between expression/statement in the syntax

- Only expressions

❑ Usual constructs:

- conditional

```
if i = 1 then 2 else 3
```

- for loop

```
for i = 1 to 10 do x := !x + i done
```

- sequence

```
x := 1; 2 * !x
```

“let = in” Expression

□ `let x = e1 in e2` is an expression

- its type and value are those of `e2`,
- in an environment where `x` has the type and value of `e1`

□ Example:

```
let x = 1 in (let y = 2 in x + y) * (let z = 3 in x * z)
```

□ Local variables:

- In C or Java, local variables appear in a block:
- In OCaml, a local variable is introduced with `let in` as for a global variable:
 - Necessarily initialized, type inferred, immutable,
 - But `scope limited to the expression following in`

```
{  
  int x = 1;  
  ...  
}
```

Programs

□ **program** = sequence of **declarations** and **expressions** to evaluate

- variables introduced with **let** and immutable
- no distinction expression / statement

```
let x = 1 + 2;;  
print_int x;;  
let y = x * x;;  
print_int y;;
```

□ Examples

Java

```
{ int x = 1;  
  x = x + 1;  
  int y = x * x;  
  System.out.print(y); }
```

OCaml

```
let x = ref 1 in  
x := !x + 1;  
let y = !x * !x in print_int y
```

unit type

□ Expressions with no meaningful value have type unit

- Expressions include assignment, loop, ...
- This type has a single value, written `()`

□ Example:

- It is the type given to the **else** branch when it is omitted
- Correct:

```
if !x > 0 then x := 0
```

- Incorrect:

```
2 + (if ! x > 0 then 1)
```

Interactive Loop

❑ **Interactive** version of the compiler: ocaml, **utop** (preferred)

```
[→ ~ ocaml
OCaml version 5.4.1
Enter #help;; for help.

# let x = 1 in x + 2;;
- : int = 3
# let y = 1 + 2;;
val y : int = 3
# y * y;;
- : int = 9
#
```

```
Welcome to utop version 2.17.0 (using OCaml version 5.4.1)!

Type #utop_help for help about using utop.

-( 18:06:13 )-< command 0 >-----{ counter: 0 }-
utop # let x = 1 in x + 2;;
- : int = 3
-( 18:06:13 )-< command 1 >-----{ counter: 0 }-
utop # let y = 1 + 2;;
val y : int = 3
-( 18:06:25 )-< command 2 >-----{ counter: 0 }-
utop # y*y;;
- : int = 9
-( 18:06:32 )-< command 3 >-----{ counter: 0 }-
utop #
```

Arg	Array	ArrayLabels	Assert_failure	Atomic	Bigarray	Bool	Buffer	Bytes	Byte
-----	-------	-------------	----------------	--------	----------	------	--------	-------	------

Function syntax

❑ Example:

```
# let f x = x * x;;  
val f : int -> int = <fun>  
# f 4;;  
- : int = 16
```

- Body = expression (no **return**)
- Type is inferred (types of argument x and result)

❑ Java vs OCaml

Java	OCaml
<pre>static int f(int x) { return x * x; }</pre>	<pre>let f x = x * x</pre>

Procedure

□ a procedure = a function whose result type is **unit**

□ Example

```
# let x = ref 1;;  
# let set v = x := v;;
```

```
val set : int -> unit = <fun>
```

```
# set 3;;
```

```
- : unit = ()
```

```
# !x;;
```

```
- : int = 3
```

Function with/without arguments

❑ Function without arguments:

- Takes an argument of type **unit**

```
# let reset () = x := 0;;
```

```
val reset : unit -> unit = <fun>
```

```
# reset ();;
```

❑ Function with several arguments

```
# let f x y z = if x > 0 then y + x else z - x;;
```

```
val f : int -> int -> int -> int = <fun>
```

```
# f 1 2 3;;
```

```
- : int = 3
```

Local function

❑ Function local to an expression

```
# let sqr x = x * x in sqr 3 + sqr 4 = sqr 5;;
```

```
- : bool -> true
```

❑ Function local to another function

```
# let pythagorean x y z =  
  let sqr n = n * n in  
  sqr x + sqr y = sqr z;;
```

```
- : bool -> true
```

Function as first-class citizen

❑ function = yet another expression, introduced with **fun**

```
# fun x -> x + 1
```

```
- : int -> int = <fun>
```

```
# (fun x -> x + 1) 3;;
```

```
- : int = 4
```

❑ Internally,

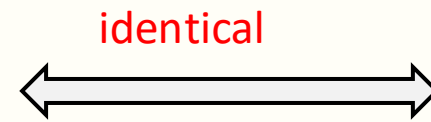
```
let f x = x + 1;;
```

is identical to

```
let f = fun x -> x + 1;;
```

Partial application

```
fun x y -> x*x + y*y;;
```



```
fun x -> fun y -> x*x + y*y;;
```

❑ one can apply a function **partially**. Example:

```
# let f x y = x*x + y*y;;
```

```
val f : int -> int -> int = <fun>
```

```
# g 4;;
```

```
# let g = f 3;;
```

```
val g : int -> int = <fun>
```

```
- : int = 25
```

❑ a partial application is a way to return a **function**

❑ but one can also return a function as the result of a computation

```
# let f x = let x2 = x * x in fun y -> x2 + y * y;;
```

```
val f : int -> int -> int = <fun>
```

● A partial application of f computes $x*x$ **only once**

Partial application: example

```
# let count_from n =  
    let r = ref (n - 1) in fun () -> incr r; !r;;
```

```
val count_from : int -> unit -> int = <fun>
```

```
# let c = count_from 0;;
```

```
val c : unit -> int = <fun>
```

```
# c ();;
```

```
- : int = 0
```

```
# c ();;
```

```
- : int = 1
```

Higher-order functions

□ A function may take functions as arguments

```
# let integral f =  
  let n = 100 in  
  let s = ref 0.0 in  
  for i = 0 to n-1 do  
    let x = float i /. float n in s := !s +. f x  
  done;  
  !s /. float n
```

```
# integral sin;;
```

```
- : float = 0.455486508387318301
```

```
# integral (fun x -> x*.x);;
```

```
- : float = 0.32835
```

Iteration

- ❑ In Java, one iterates over a collection with a cursor

```
for (Elt x: s) {  
    ... do something with x ...  
}
```

- ❑ In OCaml, we typically write

```
iter (fun x -> ... do something with x ...) s
```

- where iter is a function provided with the data structure, with type

```
val iter: (element_type -> unit) -> set -> unit
```

- Example:

```
iter (fun x -> Printf.printf "%s\n" x) s
```

Difference w.r.t. to function pointers

- ❑ In C, one can pass and return function pointers
- ❑ But OCaml functions are more than function pointers

```
let f x = let x2 = x * x in fun y -> x2 + y * y;;
```

- ❑ the value of x2 is captured in a **closure**
 - closure = function + environment

- ❑ Note: there are closures in Java (≥ 8) too

```
s.forEach(x -> { System.out.println(x); });
```

Polymorphism

```
# let f x = x;;
```

```
val f : 'a -> 'a = <fun>
```

```
# f 3;;
```

```
- : int = 3
```

```
# f true;;
```

```
- : bool = true
```

```
# f print_int;;
```

```
- : int -> unit = <fun>
```

```
# f print_int 1;;
```

```
1- : unit = ()
```

❑ OCaml always infers **the most general type**

❑ Example:

```
# let compose f g = fun x -> f (g x);;
```

```
val compose : ('a -> 'b) -> ('c -> 'a) -> 'c -> 'b = <fun>
```

Arrays

```
# let a = Array.make 10 0;;
```

```
val a : int array = [|0; 0; 0; 0; 0; 0; 0; 0; 0; 0|]
```

Necessarily initialized

```
# let a = [| 1; 2; 3; 4 |];;
```

```
# a.(1);;
```

```
- : int = 2
```

```
# a.(1) <- 5;;
```

```
- : unit = ()
```

```
# a;;
```

```
- : int array = [|1; 5; 3; 4|]
```

Memory Allocation

❑ Memory allocation handled by a **garbage collector** (GC)

❑ GC benefits:

- unused memory is reclaimed automatically
- efficient allocation

❑ Forget about “dynamic allocation is expensive”

- but keep worrying about complexity!

Java

```
int[] a = new int[42];  
a[17]  
a[7] = 3;  
a.length
```

OCaml

```
let a = Array.make 42 0  
  
a.(17)  
a.(7) <- 3  
Array.length a
```

Recursive functions

❑ In OCaml, it is idiomatic to use recursive functions

- a function call is cheap, and tail calls are optimized
- Recursive code: clearer, simpler to justify

❑ Example: Insertion Sort

```
let zero f =  
  let rec lookup i =  
    if f i = 0 then i  
    else lookup (i+1) in  
  lookup 0
```

```
let insertion_sort a =  
  let swap i j =  
    let t = a.(i) in a.(i) <- a.(j); a.(j) <- t  
  in  
  for i = 1 to Array.length a - 1 do  
    (* insert element a[i] in a[0..i-1] *)  
    let j = ref (i - 1) in  
    while !j >= 0 && a.(!j) > a.(!j + 1) do  
      swap !j (!j + 1); decr j  
    done  
  done
```

```
let insertion_sort a =  
  let swap i j =  
    let t = a.(i) in a.(i) <- a.(j); a.(j) <- t  
  in  
  for i = 1 to Array.length a - 1 do  
    (* insert element a[i] in a[0..i-1] *)  
    let rec insert j =  
      if j >= 0 && a.(j) > a.(j+1) then  
        begin swap j (j+1); insert (j-1) end  
    in  
    insert (i-1)  
  done
```

Records

- ❑ Like in most programming languages, a record type is first declared

```
type complex = { re : float; im : float }
```

- ❑ Allocation and initialization are simultaneous:

```
# let x = { re = 1.0; im = -1.0 };;
```

```
val x : complex = {re = 1.; im = -1.}
```

```
# x.im;;
```

```
- : float = -1.
```

Mutable fields

```
type person = { name : string; mutable age : int }
```

```
# let p = { name = "Martin"; age = 23 };;
```

```
val p : person = {name = "Martin"; age = 23}
```

```
# p.age <- p.age + 1;;
```

```
- : unit = ()
```

```
# p.age;;
```

```
- : int = 24
```

❑ Revisit references: a reference = a record of that predefined type

```
type 'a ref = { mutable contents : 'a }
```

- **ref**, **!** and **:=** are syntactic sugar
- only arrays and **mutable** fields can be mutated

Class in Java vs Record in OCaml

Java	OCaml
<pre>class T { final int v; boolean b; T(int v, boolean b) { this.v = v; this.b = b; } }</pre> <pre>T r = new T(42, true); r.b = false; r.v</pre>	<pre>type t = { v: int; mutable b: bool; }</pre> <pre>Let r = { v = 42; b = true } r.b <- false r.v</pre>

Tuples

□ Usual notation

```
# (1,2,3);;
```

```
- : int * int * int = (1, 2, 3)
```

```
# let v = (1, true, "hello", 'a');
```

```
val v : int * bool * string * char =  
  (1, true, "hello", 'a')
```

□ Access to components

```
# let (a,b,c,d) = v;;
```

```
val a : int = 1  
val b : bool = true  
val c : string = "hello"  
val d : char = 'a'
```

Tuples

❑ Useful to return several values

```
# let rec division n m =  
    if n < m then (0, n)  
    else let (q,r) = division (n - m) m in (q + 1, r);;
```

```
val division : int -> int -> int * int = <fun>
```

❑ Function taking a tuple as argument

```
# let f (x,y) = x + y;;
```

```
val f : int * int -> int = <fun>
```

```
# f (1,2);;
```

```
- : int = 3
```

Lists

- ❑ Predefined type of lists, α list, immutable and homogeneous built from the empty list `[]` and addition in front of a list `::`

```
# let l = 1 :: 2 :: 3 :: [];;
```

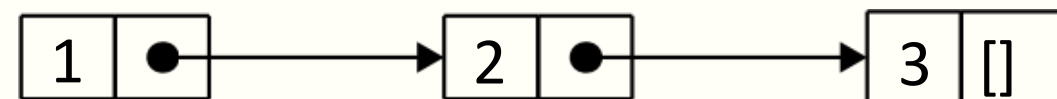
```
val l : int list = [1; 2; 3]
```

- Shorter Syntax

```
# let l = [1; 2; 3];;
```

- ❑ Presentation in memory

- Ocaml lists = identical to lists in C or Java
- The list `[1; 2; 3]` is represented as



Algebraic Data Types (ADT)

- ❑ Lists = particular case of **algebraic data type**
- ❑ Algebraic data type = union of several **constructors**

```
type fmla = True | False | And of fmla * fmla
```

```
# True;;
```

```
- : fmla = True
```

```
# And (True, False);;
```

```
- : fmla = And (True, False)
```

- ❑ Lists predefined as:

```
type 'a list = [] | :: of 'a * 'a list
```

Pattern matching

□ pattern matching = case analysis on a list

```
# let rec sum l =  
  match l with  
  | [] -> 0  
  | x :: r -> x + sum r;;  
  
val sum : int list -> int = <fun>  
  
# sum [1;2;3];;  
  
- : int = 6
```

● shorter notation for a function performing pattern matching on its argument

```
let rec sum = function  
  | [] -> 0  
  | x :: r -> x + sum r;;
```

Pattern matching

□ Pattern matching generalizes to algebraic data types

```
#let rec eval = function
  | True  -> true
  | False -> false
  | And (f1, f2) -> eval f1 && eval f2;;

val eval : fmla -> bool = <fun>
```

□ Patterns can be **nested**, **omitted** or **grouped**

```
let rec eval = function
  | True  -> true
  | False -> false
  | And (False, f2) -> false
  | And (f1, False) -> false
  | And (f1, f2) -> eval f1 && eval f2;;
```

```
let rec eval = function
  | True  -> true
  | False -> false
  | And (False, _) | And (_, False) -> false
  | And (f1, f2) -> eval f1 && eval f2;;
```

□ pattern matching is not limited to algebraic data types

- one may write **let pattern = expression** when there is a single pattern (as in let (a,b,c,d) = v for instance)

Inheritance in Java vs ADT in OCaml

Java

```
abstract class Fmla { }
class True extends Fmla { }
class False extends Fmla { }
class And extends Fmla {
    Fmla f1, f2; }

abstract class Fmla {
    abstract boolean eval(); }
class True { boolean eval() {
    return true; } }
class False { boolean eval() {
    return false; } }
class And { boolean eval() {
    return f1.eval() && f2.eval();
} }
```

OCaml

```
type fmla =
| True
| False
| And of fmla * fmla

let rec eval = function
| True -> true
| False -> false
| And (f1, f2) ->
    eval f1 && eval f2
```

Recap

- ❑ Allocation is cheap
- ❑ Memory is reclaimed automatically
- ❑ Allocated values are necessarily initialized
- ❑ Most values **cannot** be mutated
 - only arrays and mutable record fields can be
- ❑ Efficient representation of values
- ❑ Pattern matching = case analysis over values

Execution model

- ❑ A **value**, which fits on 64 bits, is
 - either a *primitive value* (integer, floating point, Boolean, [], etc.)
 - or a *pointer* (to an array, a constructor such as And, etc.)
- ❑ In OCaml, there is no **null** value
 - Any value is necessarily initialized
 - no such thing as NullPointerException in Java
 - neither segmentation fault as in C/C++
 - sometimes a pain, but it's worth the effort:
 - an expression of type τ whose evaluation terminates necessarily has a legal value of type τ
 - This is known as **strong typing**
- ❑ In OCaml, passing mode is **by value**
 - In particular, no value is ever copied
 - It is **exactly as in Java**

Comparison

□ equality written `==` is **physical equality**,

● that is, equality of pointers or primitive values as in Java

```
# (1, 2) == (1, 2);;
```

```
- : bool = false
```

□ Equality written `=`, on the contrary, is **structural equality**,

● that is, recursive equality descending in sub-terms

```
# (1, 2) = (1, 2);;
```

```
- : bool = true
```

● it is equivalent to `equals` in Java (when suitably defined)

Exceptions

- ❑ An exception may be **raised**

```
let division n m =  
  if m = 0 then raise Division_by_zero else ...
```

- ❑ and later caught

```
try division x y with Division_by_zero -> (0,0)
```

- ❑ One can introduce new exceptions

```
exception Error  
exception ParseError of string
```

- ❑ In OCaml, exceptions are used in the library to signal exceptional behavior

- example: Not found to signal a missing value, where Java typically returns null

```
try let v = Hashtbl.find table key in ...  
with Not_found -> ...
```

Modules

❑ In software engineering, when programs get big we need to

- split code into units (**modularity**)
- hide data representation (**encapsulation**)
- avoid duplicating code

❑ In OCaml, this is provided by **modules**

- In OCaml, each **file** is a **module**
- if arith.ml contains

```
let pi = 3.141592  
let round x = floor (x +. 0.5)
```

- we can use it within another module main.ml:

```
let x = float_of_string (read_line ());;  
print_float (Arith.round (x /. Arith.pi));;  
print_newline ();;
```

Encapsulation

❑ We can limit what is exported with an **interface**

● in file **arith.mli**

```
val round : float -> float
```

```
% ocamlpt -c arith.mli
```

```
% ocamlpt -c arith.ml
```

● **Arith.pi** cannot be found as it is not exported in the interface

❑ An interface may also hide the **definition** of a type

set.ml

```
type t = int list
let empty = []
let add x l = x :: l
let mem = List.mem
```

set.mli

```
type t
val empty : t
val add : int -> t -> t
val mem : int -> t -> bool
```

type **t** is an **abstract type**

Separate compilation

- ❑ Compilation of a file only depends on the **interfaces** of the other files
 - **fewer recompilation** when a code changes but its interface does not

Module System

□ We can also use keyword **module** to define a module

```
module M = struct
  let c = 100
  let f x = c * x
end
```

```
module A = struct
  module B = struct
    let f x = 6 * x
  end
  let f x = B.f (x + 1)
end
```

□ We also define **interfaces** for modules

```
module type S = sig
  val f : int -> int
end
```

```
module M : S = struct
  let a = 2
  let f x = a * x
end
```

Interface constraint

M.a is hidden

□ Keyword **open** for namespace import: “open Arith” allows use **pi** directly

Functors

❑ functor = **module parameterized** with other modules

❑ Example: Hash Table

```
module HashTable(X: HashedType) = struct

  type t = X.elm list array

  let create n = Array.make n []

  let add t x = let i = (X.hash x) mod (Array.length t) in
    t.(i) <- x :: t.(i)

  let mem t x = let i = (X.hash x) mod (Array.length t) in
    List.exists (X.eq x) t.(i)

end
```

inside, X is used as any regular module

```
module type HashedType = sig
  type elm
  val hash: elm -> int
  val eq: elm -> elm -> bool
end
```

```
module HashTable(X: HashedType) : sig
  type t
  val create : int -> t
  val add : t -> X.elm -> unit
  val mem : t -> X.elm -> bool
end
```

functor type

Functor

❑ Functor use

```
module Int = struct
  type elt = int
  let hash x = abs x
  let eq x y = x=y
end
```

```
module Hint = HashTable(Int)
```

```
# let t = Hint.create 17;;
```

```
val t : Hint.t = <abstr>
```

```
# Hint.add t 13;;
```

```
- : unit = ()
```

```
# Hint.add t 173;;
```

```
- : unit = ()
```

❑ Applications of functors

- data structures parameterized with other data structures

- Hashtbl.Make : hash tables

- Set.Make : finite sets via balanced trees

- Map.Make : finite maps via balanced trees

- algorithms parameterized with data structures

- Example: Dijkstra's algorithm

```
module Dijkstra (G: sig
  type graph
  type vertex
  val succ: graph -> vertex -> (vertex * float) list
end) : sig
  val shortest_path: G.graph -> G.vertex ->
    G.vertex -> G.vertex list * float
end
```

HashTable implementation in Java/OCaml

Java

```
interface HashedType<T> {  
  
    int hash();  
    boolean eq(T x);  
}  
  
class HashTable  
    <E extends HashedType<E>> { ...
```

OCaml

```
module type HashedType = sig  
    type elt  
    val hash: elt -> int  
    val eq: elt -> elt -> bool  
end  
  
module HashTable(E: HashedType) = struct  
    ...
```

Persistence

❑ Immutable data structures

- a value is not modified by an operation,
- but a **new** value is returned
- this is called **applicative programming** or **functional programming**

❑ In OCaml, most data structures are **immutable**

- exceptions are arrays and records with mutable fields

❑ Example of immutable structure: lists

```
let l = [1; 2; 3]
```

```
let l' = 0 :: l
```

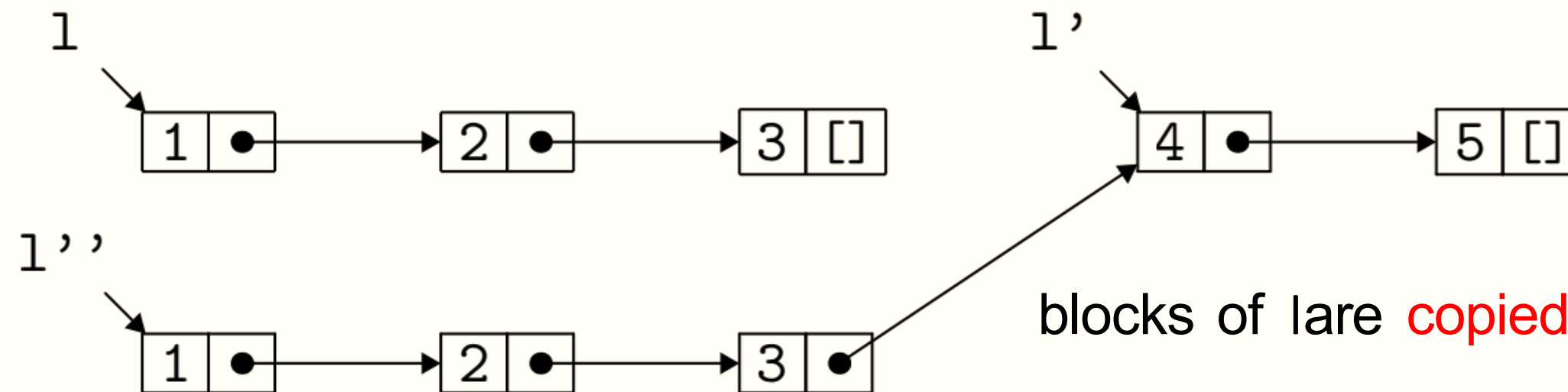


Example of immutable structure: lists

❑ Concatenating two lists:

```
let rec append l1 l2 = match l1 with  
  | [] -> l2  
  | x :: l -> x :: append l l2
```

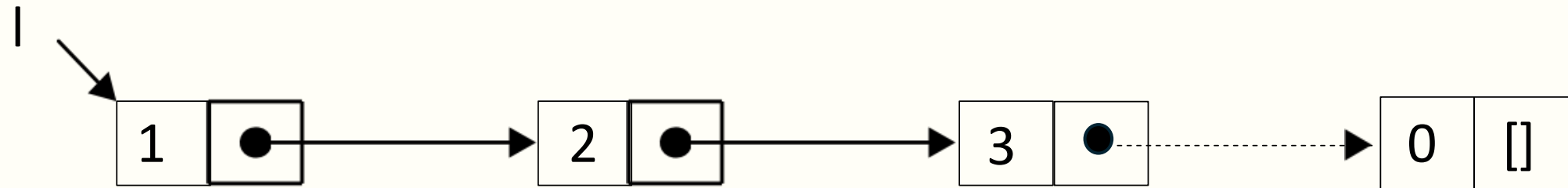
```
let l = [1; 2; 3]  
let l' = [4; 5]  
let l'' = append l l'
```



blocks of l are **copied**, blocks of l' are **shared**

Example of immutable structure: lists

❑ Counterpart: adding an element at the end of the list is not simple:



❑ Mutable linked lists

● One can implement traditional linked lists, for instance with

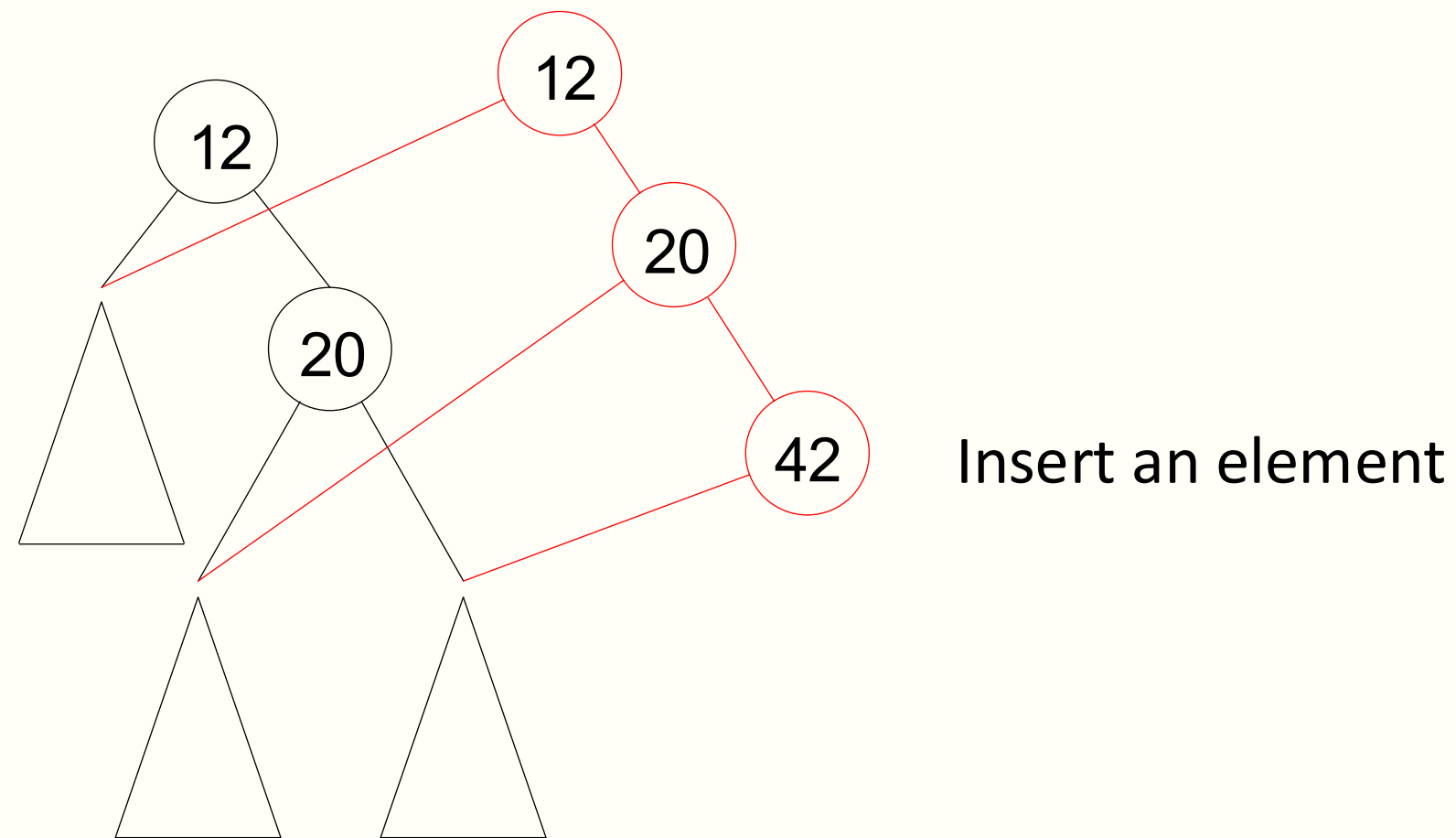
```
type 'a mlist = Empty | Element of 'a element  
and 'a element = { value: 'a; mutable next: 'a mlist }
```

● but then be careful with **sharing** (aliasing)

Example of immutable structure: trees

```
type tree = Empty | Node of int * tree * tree
```

```
val add : int -> tree -> tree
```



Again, few copies and mostly sharing!

Persistence and Backtracking

□ Example A:

- search for a path in a maze

```
type state
val is_exit : state -> bool
type move
val moves : state -> move list
val move : state -> move -> state
```

```
let rec search e =
  is_exit e || iter e (moves e)
and iter e = function
  | [] -> false
  | d :: r -> search (move e d)
               || iter e r
```

- without persistence, with a mutable, global state

```
let rec search () =
  is_exit () || iter (moves ())
and iter = function
  | [] -> false
  | d :: r -> (move d; search ()) || (undo d; iter r)
```

i.e. one has to **undo** the side effect
(here with a function undo, inverse
of move)

Persistence and Backtracking: Example B

□ Simple Java fragments, represented with

```
type stmt =  
  | Return of string  
  | Var   of string * int  
  | If    of string * string *  
          stmt list * stmt list
```

```
int x = 1;  
int z = 2;  
if (x == z) {  
  int y = 2;  
  if (y == z) return y;  
  else return z;  
} else  
  return x;
```

Example code

● let us check that any variable which is used was previously declared (within a list of statements)

```
let rec check_instr vars = function  
  | Return x -> List.mem x vars  
  | If (x, y, p1, p2) -> List.mem x vars && List.mem y vars &&  
    check_prog vars p1 && check_prog vars p2  
  | Var _ -> true  
and check_prog vars = function  
  | [] -> true  
  | Var (x, _) :: p -> check_prog (x :: vars) p  
  | i :: p -> check_instr vars i && check_prog vars p
```

Persistence and Backtracking: Example C

- ❑ a program handles a database
 - non atomic updates, requiring lot of computation
- ❑ with a mutable state

```
try
  ... performs update on the database ...
with e ->
  ... rollback database to a consistent state ...
  ... handle the error ...
```

- ❑ with a persistent data structure

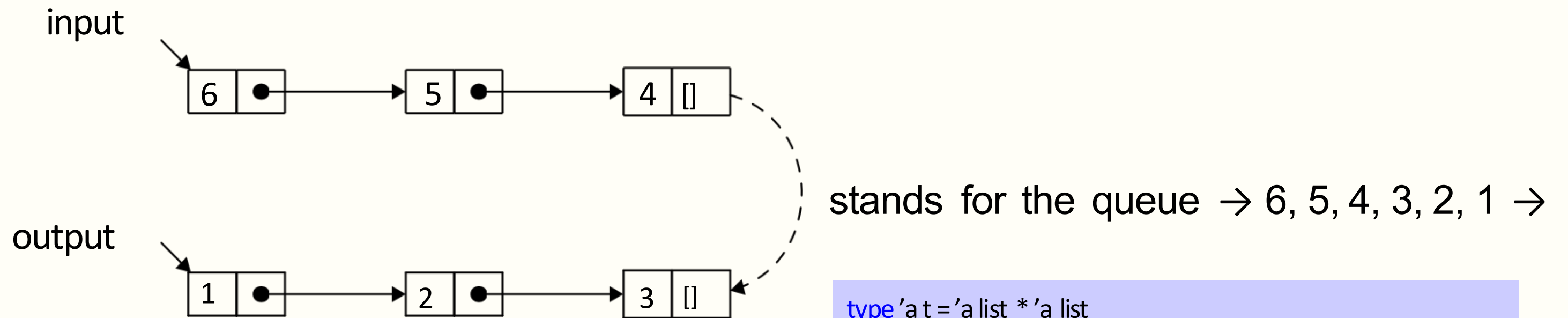
```
let bd = ref (... initial database ...)

...

try
  bd := (... compute the update of !bd ...)
with e ->
  ... handle the error ...
```

Example of immutable structure: persistent queues

❑ Idea: a queue is a **pair of lists**, one for insertion, and one for extraction



❑ Define **persistent queue**

```
type 'a t
val create : unit -> 'a t
val push : 'a -> 'a t -> 'a t
exception Empty
val pop : 'a t -> 'a * 'a t
```

Implementation



```
type 'a t = 'a list * 'a list
let create () = [], []
let push x (e,s) = (x :: e, s)
exception Empty
let pop = function
  | e, x :: s -> x, (e,s)
  | e, [] -> match List.rev e with
    | x :: s -> x, ([], s)
    | [] -> raise Empty
```

Example of immutable structure: persistent queues

□ When accessing several times the same queue whose second list is empty, we reverse several times the same list

● let's add a reference to register the list reversal the first time it is performed

```
type 'a t = ('a list * 'a list) ref
```

● the side effect is done “under the hood”, in a way not observable from the user, the contents of the queue staying the same

□ persistent = observationally immutable

● Persistence does not mean absence of side effects

● immutable → persistent

● the reciprocal is wrong

```
type 'a t = 'a list * 'a list
let create () = [], []
let push x (e,s) = (x :: e, s)
exception Empty
let pop = function
  | e, x :: s -> x, (e,s)
  | e, [] -> match List.rev e with
    | x :: s -> x, ([], s)
    | [] -> raise Empty
```

Benefits of Persistence

□ persistent structure = no observable modification

- In OCaml: List, Set, Map
- can be very efficient (lot of sharing, hidden side effects, no copies)
- idea independent of OCaml

□ Benefits of Persistence

- **Correctness** of programs
 - code is simpler
 - mathematical reasoning is possible
- Easy to perform **backtracking**
 - search algorithms
 - symbolic manipulation and scopes error recovery

Ocaml Tools

□ opam

package manager

□ utop

a more fully-featured interactive top-level

~~● ocaml~~

~~the top-level interactive loop~~

□ ocamlc

the bytecode compiler

□ ocamlc.opt

the native code compiler

□ ocamllex

the lexer generator

□ menhir

a more modern parser generator

~~● ocaml yacc~~

~~the parser generator~~

□ dune

build system

~~● ocamlbuild~~

~~a compilation manager~~

References

- ❑ A Short Introduction to OCaml, Jean-Christophe Filli^atre, <https://www.enseignement.polytechnique.fr/profs/informatique/Jean-Christophe.Filliatre/14-15/INF549/slides.pdf>
- ❑ OCaml Documentation, <https://ocaml.org/docs>

关于Ocaml语言的任何问题， 多问问AI大模型！也欢迎在群里讨论交流

作业

□1. List Processing

给定成绩列表如 : `let scores = [45; 82; 60; 92; 39]`, 使用`List.map`, `List.filter`, `List.fold_left`等完成如下函数 :

```
val passed : int list -> int list
val add_bonus : int -> int list -> int list
val average : int list -> float
```

□2. Expression Evaluator

定义如下类型 :

```
type expr = Int of int | Var of string | Add of expr * expr | Let of string * expr * expr
```

```
type env = (string * int) list
```

实现如下函数 :

```
val lookup : string -> env -> int option
```

```
val eval : env -> expr -> int option
```

满足样例 :

```
eval [] (Add (Int 3, Int 4)) (* => Some 7 *)
```

```
eval [] (Let ("x", Int 10, Add (Var "x", Int 5))) (* => Some 15 *)
```

```
eval [] ( Let ("x", Int 10, Let ("x", Int 20, Add (Var "x", Int 1))) ) (* => Some 21 *)
```